

Fig. 2 shows a prototype of an ESP32-based smart power monitoring and control system assembled

on a breadboard. It consists of an ESP32 development board, a power supply module, a step-down

transformer, and supporting components interconnected with jumper wires. The setup is designed for

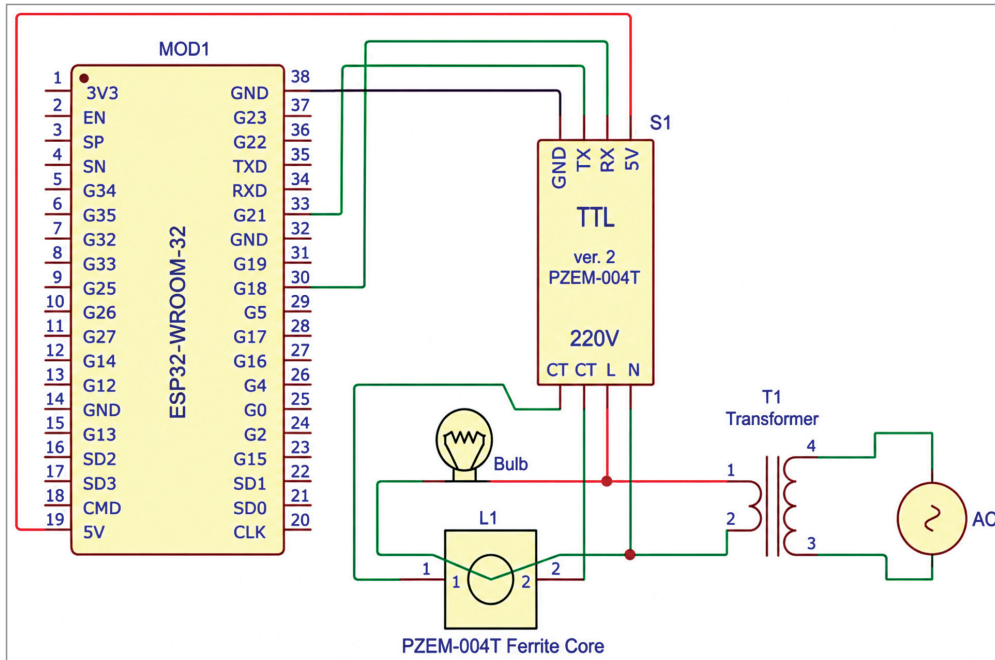


Fig. 4: Circuit diagram

```

1 import firebase_admin
2 from firebase_admin import credentials, db
3 from openpyxl import Workbook, load_workbook
4 from datetime import datetime
5 import numpy as np
6 import os
7 import time
8
9 from sklearn.preprocessing import MinMaxScaler
10 from tensorflow.keras.models import Sequential
11 from tensorflow.keras.layers import LSTM, Dense, Dropout
12 from tensorflow.keras import Input
13
14 # ===== CONFIG =====
15
16 cred_path = r"F:\sherin_predict\serviceAccountKey.json"
17 excel_file = r"F:\sherin_predict\SmartGrid_Data.xlsx"
18 OVERLOAD_LIMIT = 1500 # mW
19
20 # ===== FIREBASE =====
21
22 if not firebase_admin._apps:
23     cred = credentials.Certificate(cred_path)
24     firebase_admin.initialize_app(cred, {
25         'databaseURL': "https://load-prediction-e14b6-default-rtdb.asia-southeast1.firebaseio.com/"
26     })
27     print("Firebase Connected")
28
29 gen_ref = db.reference('/SmartGrid/Live')
30 pred_ref = db.reference('/SmartGrid/Prediction')
31
32 # ===== EXCEL =====
33
34 def init_excel():
35     if not os.path.exists(excel_file):
36         wb = Workbook()
37         ws = wb.active
38         ws.title = "RealTimeData"
39         ws.append([
40             "Timestamp", "Voltage", "Current_mA", "TotalPower_mW",
41             "Frequency", "PowerFactor", "5min_Pred", "Daily_Pred",
42             "Yearly_Pred", "Daily_Peak", "Monthly_Peak", "Yearly_Peak"
43         ])

```

Fig. 5: Python code snippet

testing data acquisition, sensor interfacing, wireless communication, and real-time monitoring before final PCB fabrication. Such prototypes help verify circuit operation, troubleshoot hardware, and validate system performance during development.

Fig. 3 shows the block diagram of the proposed AI-based substation load monitoring and prediction system. The AC supply is measured through voltage and current sensing, and the data is acquired by the ESP32 microcontroller. The components required to build this system are

listed in Bill of Materials table.

Circuit diagram and working

Fig. 4 shows the circuit diagram of the load monitoring and prediction system. It is built around an ESP32-WROOM-32, a PZEM-004T energy monitoring module, a 230V-to-24V, 1A step-down transformer, and a 60W bulb used as the load. The transformer provides an isolated AC voltage to the PZEM-004T for voltage measurement,